

#1. $\vec{r} = (0.5 + u)\vec{s}_1$, $\vec{w} = \vec{s}_3$

A. Let assume that the spacecraft is in equilibrium, then $\ddot{x} = \ddot{y} = \ddot{z} = 0$.

$$(i) \ddot{x} - 2\ddot{y} - x = -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3}$$

- Left side: $\ddot{x} - 2\ddot{y} - x = -x = -u + 0.5$

- Right side: $|r_1| = 0.5 = |r_2|$

$$\Rightarrow -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3} = +\frac{(1-u)0.5}{0.5^3} - \frac{0.5u}{0.5^3}$$

$$= 4(1-u-u) = 4(1-2u)$$

Then $-u + 0.5 = 4(1-2u) \Rightarrow u = 0.5$. Since $u \neq 0.5$, then the left side and the right side are different

$$(ii) \ddot{y} + 2\ddot{x} - y = -\frac{(1-u)y}{r_1^3} - \frac{uy}{r_2^3}$$

- Left side: $\ddot{y} + 2\ddot{x} - y = 0$

- Right side: Since $y = 0$, 0.

Thus the left and the right side are same.

$$(iii) \ddot{z} = -\frac{(1-u)z}{r_1^3} - \frac{uz}{r_2^3} \Rightarrow \text{both sides are zero.}$$

Therefore, the spacecraft is not in equilibrium

There is only S_1 -axis acceleration.

$$\Rightarrow \ddot{x} - u + 0.5 = 4(1-2u)$$

$$\Rightarrow \ddot{x} = 7(0.5 - u)$$

Thus, the acceleration $\ddot{\vec{r}} = 7(0.5 - u)\vec{s}_1$.

B. For keeping the spacecraft in equilibrium at the midway point, an acceleration of the spacecraft should be zero.

Since only S_1 -coordinate component is not in equilibrium, it is enough to control only S_1 -coordinate component

$$\ddot{x} - 2\ddot{y} - x = -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3} + u$$

$$\Rightarrow -u + 0.5 = 4(1-2u) + u$$

$$\Rightarrow u = 7(u - 0.5)$$

Hence the control acceleration vector is $\vec{U} = 7(u - 0.5)\vec{s}_1$.

#2. Let $R = R_x(\phi)R_y(\theta)R_z(\psi)$

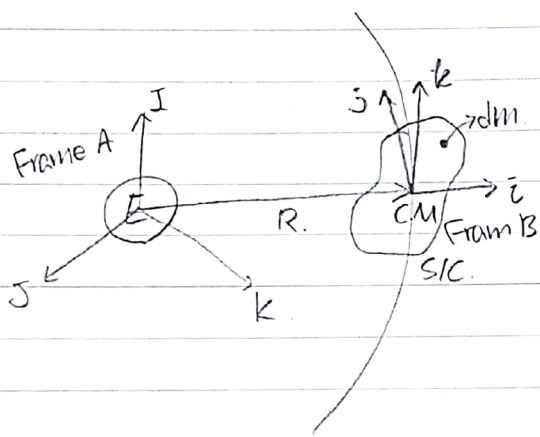
$$\Rightarrow \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \dot{\vec{\phi}} + \vec{\omega} + \vec{\dot{\psi}} = R_x(\phi)R_y(\theta) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + R_x(\phi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned}
 &= \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ \sin\phi \sin\theta & \cos\phi & \sin\phi \cos\theta \\ \cos\phi \sin\theta & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & \sin\theta \\ 0 & \cos\phi & \sin\phi \cos\theta \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \\
 \therefore \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} &= \begin{bmatrix} 1 & 0 & \sin\theta \\ 0 & \cos\phi & \sin\phi \cos\theta \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}
 \end{aligned}$$

#3

A. $I = \text{diag}(A, B, C)$, $\vec{M} = [M_1, M_2, M_3]^T$, $\vec{\omega} = [\omega_1, \omega_2, \omega_3]^T$



$$\begin{aligned}
 H &= \int_B \vec{r} \times \vec{v} \, dm = \int_B \vec{r} \times (\vec{\omega} \times \vec{r}) \, dm \\
 &= \int_B (r^T r - r r^T) \vec{\omega} \, dm \\
 &= \int_B (r^T r - r r^T) \, dm \vec{\omega} \\
 &= \int_B \begin{bmatrix} r_y^2 + r_z^2 & -r_x r_y & -r_x r_z \\ -r_x r_y & r_x^2 + r_z^2 & -r_y r_z \\ -r_x r_z & -r_y r_z & r_x^2 + r_y^2 \end{bmatrix} \, dm \vec{\omega} \\
 &= I \vec{\omega}.
 \end{aligned}$$

By Coriolis's theorem, $\left(\frac{d}{dt} \vec{H}\right)^A = \left(\frac{d}{dt} \vec{H}\right)^B + \vec{\omega} \times \vec{H}$.

$$\Rightarrow I \dot{\vec{\omega}} = \vec{\omega} \times \vec{H} + \vec{M} = \vec{\omega} \times I \vec{\omega} + \vec{M}.$$

$$\therefore I \dot{\vec{\omega}} - \vec{\omega} \times I \vec{\omega} = \vec{M}.$$

Since $I = \text{diag}(A, B, C)$,

$$\begin{cases} A \dot{\omega}_1 = (B - C) \omega_2 \omega_3 + M_1 \\ B \dot{\omega}_2 = (C - A) \omega_1 \omega_3 + M_2 \\ C \dot{\omega}_3 = (A - B) \omega_1 \omega_2 + M_3 \end{cases}$$

B. Let $\vec{M} = 0$, then the kinetic energy $T = \frac{1}{2} \vec{\omega}^T I \vec{\omega}$

$$\frac{d}{dt} T = \frac{d}{dt} \left(\frac{1}{2} \vec{\omega}^T I \vec{\omega} \right) = \vec{\omega}^T I \dot{\vec{\omega}} = \vec{\omega}^T (\vec{\omega} \times I \vec{\omega})$$

Since $a \cdot (b \times c) = \det([a, b, c]^T)$,

$$\frac{d}{dt} T = (\vec{\omega}) \cdot (\vec{\omega} \times I \vec{\omega}) = \det([\vec{\omega}, \vec{\omega}, I \vec{\omega}]^T) = 0.$$

Thus T is conserved. Let $H = |H| \neq 0$.

$$\begin{aligned}
 \frac{d}{dt} H^2 &= 2 \vec{H} \cdot \dot{\vec{H}} = (I \dot{\vec{\omega}})^T (I \vec{\omega}) = (\vec{\omega} \times I \vec{\omega})^T I \vec{\omega} \\
 &= \det([I \vec{\omega}, \vec{\omega}, I \vec{\omega}]) = 0.
 \end{aligned}$$

Thus, H is conserved.

#4. $I_y \ddot{\Delta\theta} = 3\omega_0^2 (-I_x + I_z) \Delta\theta + M_y$
 $\omega_0 = 0.0012 \text{ rad/sec}$. $I = \text{diag}(150, 100, 200) \text{ kgm}^2$
A. $100 \ddot{\Delta\theta} = 3\omega_0^2 (-150 + 200) \Delta\theta$
 $\Rightarrow \ddot{\Delta\theta} = 0.03\omega_0^2 \cdot 50 \Delta\theta = 1.5\omega_0^2 \Delta\theta$
 $\Rightarrow \begin{bmatrix} \ddot{\Delta\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1.5\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \dot{\Delta\theta} \end{bmatrix}$
 $\Rightarrow \text{eigenvalues } \lambda^2 - 1.5\omega_0^2 = 0$
 $\Rightarrow \lambda = \pm 1.4697 \times 10^{-3}$

Since there exists positive eigenvalue, then this system is NOT stable.
whatever an initial state is.

B. Let $M_y = -k_p \Delta\theta - k_d \dot{\Delta\theta}$
 $\Rightarrow \ddot{\Delta\theta} = 1.5\omega_0^2 \Delta\theta - 0.01 k_p \Delta\theta - 0.01 k_d \dot{\Delta\theta}$
 $\Rightarrow \ddot{\Delta\theta} + 0.01 k_d \dot{\Delta\theta} + (0.01 k_p - 1.5\omega_0^2) \Delta\theta = 0$
 $\Rightarrow (s^2 + 0.01 k_d s + (0.01 k_p - 1.5\omega_0^2)) \Delta\theta = 0$
 $\Rightarrow s^2 + 0.01 k_d s + (0.01 k_p - 1.5\omega_0^2) = 0$
 $\Rightarrow 0.01 k_d = 0.0112 \text{ \& } 0.01 k_p = 1.5\omega_0^2 + 0.000064$
 $\Rightarrow k_d = 1.12 \text{ \& } k_p = 6.616 \times 10^{-3}$
 $\therefore M_y = -6.616 \times 10^{-3} \Delta\theta - 1.12 \dot{\Delta\theta}$

#5. Rotational matrix. R

An element of special orthogonal group $SO(3)$

Let $\{\hat{i}, \hat{j}, \hat{k}\}$ be a basis of rotating frame and $\{\hat{I}, \hat{J}, \hat{K}\}$ be a basis of a standard frame, Then R is formed by projection of $\hat{i}, \hat{j}, \hat{k}$ to $\hat{I}, \hat{J}, \hat{K}$.

$$R = \begin{bmatrix} \hat{i} \cdot \hat{I} & \hat{i} \cdot \hat{J} & \hat{i} \cdot \hat{K} \\ \hat{j} \cdot \hat{I} & \hat{j} \cdot \hat{J} & \hat{j} \cdot \hat{K} \\ \hat{k} \cdot \hat{I} & \hat{k} \cdot \hat{J} & \hat{k} \cdot \hat{K} \end{bmatrix}$$

$\exists$ 9 parameters. $\nexists$ Singularity

- Euler angles

$\exists$ 3 parameters. $\exists$ Singularity at $\theta = \pm \pi/2$.

$\exists$ 12 combinations.

Usually use 3-2-1 OR 3-1-3 combination

- Quaternion

$\exists$ 4 parameter $\nexists$ Singularity

Defined using Euler principle axis $\hat{a}$ and angle ϕ

$$\hat{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \sin \frac{\phi}{2} \hat{a} \quad q_4 = \cos \frac{\phi}{2}$$

Convenience for analysis.